

Ninad Divekar

DevOps Lead

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SUMMARY

Dual-certified DevOps Lead — AWS Solutions Architect Associate & HashiCorp Terraform Associate — with more than 5 years in IT including hands-on experience in AWS DevOps. Experienced in production-grade Kubernetes on EKS (Karpenter, Kyverno, ArgoCD, GitOps, Prometheus/Grafana), ECS Fargate/EC2, Terraform IaC, GitLab/CircleCI CI/CD, Docker, Helm, and Ansible. Delivered measurable impact: slashed infrastructure costs by 30–40%, accelerated deployments by 40%, cut build times by 40–60%, automated 200+ CloudWatch alarms, and reduced database migration time from 6+ hours to under 3 hours. Recently extended into AI/ML infrastructure — Integrated AWS Bedrock for prompt engineering (doc generation, alarm summarization). Built LangChain ReAct agent for EKS manifest analysis. Evaluated SageMaker for potential pipeline integration. Led internship programs with 80%+ project completion.

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate:** | Credly: credly.com/badges/1550ec8b-82d2-45a5-a05b-4b166967d33f
- HashiCorp Certified: Terraform Associate (003):** | Credly: credly.com/badges/78fd4d91-6019-4d2b-9eaa-651dd1b53ec4

SKILLS

Cloud & IaC: Amazon Web Services (AWS), Terraform, Cloudformation

Container Orchestration: Kubernetes (EKS), Helm, Karpenter, Kyverno, ArgoCD, Docker, AWS ECS Fargate/EC2

CI/CD & GitOps: GitLab CI/CD, GitHub Actions, CircleCI, AWS CodePipeline/CodeBuild, ArgoCD GitOps

Observability: Prometheus, Grafana, Fluent Bit, AWS CloudWatch, Nagios, ELK (Filebeat), Velero

AI/ML & Agentic: AWS Bedrock, AWS SageMaker, OpenHands, Claude API, GitHub Copilot, Prompt Engineering

Configuration & Automation: Ansible, Python, Bash Scripting, AWS Lambda

Networking & Security: AWS VPC, NGINX, Security Groups, IAM/IRSA, STS Assume Role, SSM Parameter Store, External Secrets Operator, ACM, WAF

Databases: AWS RDS (MySQL/MariaDB/Postgres), AWS DocumentDB, MongoDB, AWS DMS, Redis

Version Control & OS: Git, GitHub, GitLab, Bitbucket, Linux

Documentation: Confluence, MkDocs

EMPLOYMENT HISTORY

DevOps Lead

April 2024 – Present

Cloudshastra Infotech Solutions | Pune, India

Project: Production-Grade 3-Tier Application on AWS EKS

Key Achievements:

- EKS Cluster Provisioning:** Provisioned production-grade EKS cluster (three-tier-cluster) on AWS us-east-1, configured managed node groups, IAM OIDC, and kubeconfig access via EC2 jumpbox for secure kubectl operations
- Karpenter Node Autoscaling:** Deployed Karpenter v1.5.0 replacing Cluster Autoscaler — configured NodePool with AL2023 AMI alias, WhenEmptyOrUnderutilized consolidation policy, and preferred pod anti-affinity for intelligent cost-optimized node provisioning
- GitOps with ArgoCD:** Implemented full GitOps deployment model using ArgoCD App-of-Apps pattern — GitHub Actions CI pipeline builds, scans (Trivy), and pushes images to ECR with OIDC keyless auth; ArgoCD watches Git manifests and syncs to EKS with drift detection and automated rollback

- **Policy Enforcement with Kyverno:** Deployed Kyverno as Kubernetes-native policy engine — enforced pod security standards, image registry restrictions, resource limit requirements, and namespace-level governance across all workloads
- **External Secrets Operator:** Implemented External Secrets Operator with IRSA for each workload — MongoDB credentials, API keys, and environment configs pulled from AWS Secrets Manager at runtime; zero hardcoded secrets in any manifest
- **Observability Stack:** Deployed full observability stack using Helm — Prometheus for metrics scraping, Grafana for dashboards and alerting, Fluent Bit with FluentBitIRSARole for centralized log shipping to CloudWatch Logs; end-to-end visibility across all 3 tiers
- **Velero Backup & DR:** Configured Velero for cluster backup and disaster recovery — scheduled backups to S3 with VeleroRole IRSA, enabling point-in-time restore of Kubernetes resources and persistent volumes
- **HTTPS & DNS Automation:** Automated HTTPS using AWS ACM certificates with ExternalDNS managing Route53 records — domain app.learn.cloudshastrainfotech.com fully automated from ingress to SSL termination
- **IRSA Per Workload:** Configured individual IRSA roles per service — BackendIRSARole, ExternalSecretsRole, FluentBitRole, VeleroRole — enforcing least-privilege AWS access at the pod level
- **Multi-stage Docker Builds:** Resolved OpenSSL compatibility by upgrading react-scripts to 5.0.1 — optimized multi-stage Dockerfiles for React frontend and Node.js backend reducing final image sizes significantly

Project: AWS Infrastructure & Containerization

Key Achievements:

- **Scalable Cloud Architecture:** Designed and automated scalable AWS cloud infrastructure leveraging ECS and ECR for containerized workload management
- **Serverless Container Integration:** Implemented containerized AWS Lambda functions utilizing container image support for flexible deployment and enhanced portability
- **CI/CD Pipeline Automation:** Built end-to-end CI/CD pipelines using AWS CodeCommit, CodeBuild, and CodePipeline to ensure secure, high-availability, and efficient delivery workflows
- **Infrastructure Reliability:** Achieved consistent deployment processes across environments, reducing deployment errors and improving system availability

Project: Serverless S3 Antivirus Scanning Solution

Key Achievements:

- **Serverless Security Architecture:** Designed serverless malware scanning pipeline using AWS Lambda, Terraform, and CircleCI for automated threat detection
- **Event-Driven Scanning:** Implemented Lambda triggers on S3 uploads to initiate real-time ClamAV scans, ensuring secure and virus-free file storage with 100% scanning coverage
- **Continuous Deployment:** Integrated CircleCI for automated deployment workflows allowing easy scaling and environment replication across dev, staging, and production

Project: AI/ML Infrastructure & Agentic DevOps

Key Achievements:

- **Bedrock-Powered Alarm Intelligence:** Built automated CloudWatch alarm summarization system using Amazon Nova Micro (Bedrock Converse API) and boto3; designed prompt engineering to parse alarm JSON and generate actionable incident responses with severity scoring (P1/P2/P3) and remediation priorities; integrated into incident response runbooks, reducing mean-to-acknowledge by ~30%
- **LangChain ReAct Agent for K8s:** Developed agent that reads EKS manifests, validates against best practices (security context, resource limits, pod anti-affinity), and outputs remediation suggestions — all via Bedrock (Claude) + custom tools
- **Agentic Tooling Evaluation:** Deployed OpenHands + Cline with local Ollama/Qwen for infrastructure-as-code tasks; successfully automated deployment of 3-tier app (React + Node.js + DocumentDB) on Minikube, including troubleshooting failed pod scheduling and misconfigured services

Project: Multi-Region VPC Flow Log Automation

Key Achievements:

- **Centralized Network Monitoring:** Engineered multi-region VPC Flow Log system using Terraform, enabling centralized and scalable network traffic monitoring across AWS regions

- **CI/CD Integration:** Integrated CircleCI for continuous integration, automating deployment and validation of infrastructure updates across multiple regions
- **Regional Compliance:** Ensured compliance with network monitoring requirements across all operational regions with standardized logging configurations

Project: Internship Program Management & Infrastructure Optimization

Key Achievements:

- **Program Development:** Designed and executed structured internship program with clear roadmaps, milestones, and deliverables achieving 80%+ project completion rates
- **Cost Optimization:** Managed cost optimization across AWS accounts — achieved 25-30% cost reduction through rightsizing ECS tasks, resource auditing, and enforcing tagging for Cost Explorer visibility
- **Workflow Automation:** Developed automation scripts using Python and Bash, reducing manual workloads by 60% and improving overall workflow efficiency

Project: Automated CloudWatch Monitoring Infrastructure

Key Achievements:

- **Infrastructure Automation:** Designed and deployed Terraform modules for automated CloudWatch alarm management across ECS clusters including Services, ALB, Tasks and DocumentDB, eliminating manual configuration for 200+ monitoring alarms
- **Dynamic Resource Discovery:** Implemented dynamic service discovery logic using Terraform data sources to automatically detect and monitor new ECS services without code changes
- **Operational Efficiency:** Automated CloudWatch alarms cutting manual configuration time from 45-50 minutes to 10-15 minutes per environment

Client: InvoLead

Project: Autoflow Chatbot — Multi-Service Medical Chatbot with RAG + Guardrails

Key Achievements:

- **RAG Infrastructure:** Provisioned Qdrant vector DB + self-hosted Ollama (nomic-embed-text, 768-dim) on ECS Fargate, enabling hybrid semantic + keyword search across 50 retrieved chunks per query
- **LLM Platform:** Integrated Claude Sonnet 4 via AWS Bedrock (Converse Stream API, temp=0.0) with IRSA-scoped access, serving real-time streamed responses with autonomous tool calling for RAG search; enforced deterministic outputs (temp=0.0) + source citations for medical compliance
- **Security Hardening:** Eliminated long-lived credentials using AWS STS Assume-Role for Bedrock + S3 access
- **Guardrails Infra:** Containerized NeMo Guardrails (Llama 3.1 8B/70B via NVIDIA NGC) as sidecar service handling input PII detection and output hallucination validation before response delivery
- **Mono-Repo CI/CD Pipeline:** Managed GitLab CI/CD mono-repo pipeline supporting multiple services (frontend, backend, data_ingest, email_lambda) with folder-level change detection reducing unnecessary builds by 70%
- **Modular Infrastructure:** Automated AWS infrastructure using modular Terraform for ECR, ALB/Target Groups, ECS Task Definitions, and Fargate Services with workspace isolation
- **Network Architecture:** Configured NGINX reverse proxy within private non-routable subnets for secure internal routing and service mesh capabilities

Project: Modcon — AI Document Extraction Platform

Key Achievements:

- **AI-Powered Extraction:** Built multi-service platform using Claude 3.5 Sonnet (Bedrock) + AWS Textract to extract structured content (headers, body copy, footnotes, references) from pharmaceutical PDFs/DOCX/PPT
- **Parallel LLM Processing:** Implemented intelligent page batching (2–4 pages/batch, 80K tokens) + ThreadPoolExecutor (5 threads) reducing extraction latency by 40% for 10+ page documents
- **Prompt Engineering:** Designed chain-of-thought V4 prompts with few-shot examples + pharmaceutical glossary, enforcing source citations and verbatim extraction (temperature=0.5)
- **Smart Document Cropping:** Built PDF cropper (target_ratio=1.5) to focus on content regions, keeping multi-page docs within Claude's 200K context window
- **Microservices on ECS:** Deployed 4 services (AI, Backend, Runner, Streamlit UI) on ECS EC2 with Terraform; mono-repo GitLab CI with selective service triggers

Project: CAIS — Clinical AI Services Platform

Key Achievements:

- **Multi-Environment Infrastructure:** Designed backend infrastructure for 9 ECS Fargate-based services across 4 environments (dev/stage/uat/prod) with complete isolation
- **Hybrid Architecture Migration:** Migrated AI microservice from ECS to AWS Lambda for cost optimization, integrating back via VPC networking for seamless service communication
- **Database Synchronization:** Implemented AWS DMS for controlled DocumentDB sync between dev and prod environments ensuring data consistency for testing

DevOps Engineer

December 2021 – March 2024

Escalon Business Services | Mohali, India

Project: HUB

Key Achievements:

- **Deployment Automation:** Developed comprehensive Ansible playbooks for automated application deployment, configuration management, and environment-specific warm-up procedures — reducing deployment time by 50% and eliminating configuration drift
- **Centralized Secrets Management:** Automated parameter management using AWS SSM Parameter Store integrated with Ansible and GitLab CI/CD for .env file updates, eliminating manual configuration errors
- **Cost Optimization:** Implemented event-driven Lambda functions for automated EC2/RDS scheduling achieving 60% reduction in non-production infrastructure costs during off-hours
- **Parallel Migration Architecture:** Designed parallel database restore architecture reducing cross-region migration time from 6+ hours to under 3 hours using bash scripting and AWS CLI with intelligent mysqldump strategy
- **Infrastructure as Code:** Established IaC practices using Terraform and GitLab CI/CD for version-controlled staging infrastructure management enabling reproducible deployments

Project: RPA — Database Query Automation Platform

Key Achievements:

- **Custom Runner Infrastructure:** Built custom GitLab Runner Docker image with pre-configured database clients pushed to Amazon ECR for centralized image management
- **Containerized Database Operations:** Designed containerized database operations platform within GitLab CI/CD using custom ECR images, executing RDS queries in isolated Docker containers
- **Pipeline Automation:** Automated database maintenance workflows and data processing pipelines through GitLab CI/CD reducing manual intervention and eliminating human error

iOS Developer

Mobitrail | Mumbai, India

Key Achievements:

- Designed and developed native iOS applications using Swift and Objective-C for enterprise banking clients including ICICI ESOP and Axis Bank Banking App
- Implemented RESTful API integration, Core Data storage, SSL pinning, and Keychain Services for banking-grade security compliance
- Managed full application lifecycle from development through App Store deployment including provisioning profiles, code signing, and TestFlight beta testing

EDUCATION

PG-DMC — Advanced Computing

2016

C-DAC Institute, Mumbai (Juhu)

BE — Electronics & Telecommunication

2015

Rajendra Mane College of Engineering & Technology, Devrukh